ANSHUL KAUSHAL

AI Researcher
M.Tech. Artificial Intelligence
Indian Institute Of Technology, Patna
Focus: LLM Safety, Mechanistic Interpretability, Multimodal Learning, Computer Vision

kaushal1anshul@gmail.com
github.com/Enthusiast101
linkedin.com/in/kaushal1anshul
enthusiast101.github.io
Google Scholar

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (AI)	Indian Institute of Technology, Patna	–	2026 – Present
B.E. (CSE)	Panjab University	9.07 / 10	2022 – 2026

Summary

AI researcher with 5 research internships and 6 publications (incl. BMVC, EMNLP Demo, ApJ), focused on LLM safety, mechanistic interpretability, and multimodal representation learning, with prior work in medical image segmentation and scientific ML. Hands-on with PyTorch, Hugging Face, DINOv2, MetaCLIP, Mamba SSMs, and LLM auditing and steering.

Research Experience

- **Lexsi Labs (Aurionpro)** *Remote, Maharashtra, India*
  Research Intern – Mechanistic Interpretability and Safety in LLMs Feb 2026 – Aug 2026
  – Built **SAFETUNE**, a unified AI safety framework integrating **118+ methods** for auditing, repairing, and preserving safety in fine-tuned language models.
  – Applied **mechanistic interpretability**, safety circuit discovery, and steering vectors to analyze multilingual LLM safety and support model alignment.

- **Indian Institute of Technology Jodhpur** *Jodhpur, Rajasthan, India*
  ACM IKDD Scholar – Self-Supervised Vision and Multimodal Learning May 2025 – Jan 2026
  – Engineered a CLIP-like image-text representation model by integrating the **DINOv2** self-supervised vision encoder into **MetaCLIP** for downstream multimodal large language model (MLLM) applications.
  – Aligned the vision backbone using **99K CC and LAION image-caption pairs**, lowering data and compute requirements for multimodal representation learning.
  – Raised **zero-shot classification by approximately 2%** across ImageNet-1K, ObjectNet, and iNAT2021 benchmarks.

- **National Astronomical Observatory Japan** *Remote, Mitaka, Japan*
  Research Intern – Sparse Signal Reconstruction Nov 2024 – Nov 2025
  – Reconstructed sparse **Gamma-Ray Burst (GRB) light-curve signals** across four categories: Good, Break, Flares, and Break with Flares.
  – Benchmarked Conditional GANs, **Mamba state-space models (SSMs)**, and polynomial fitting for sparse signal reconstruction.
  – Lowered uncertainty relative to prior studies to **37% for** $\log T_a$**, 38% for** $\log F_a$**, and 41% for** α **parameters**.

- **IISER Bhopal** *Bhopal, Madhya Pradesh, India*
  Research Intern – Medical Image Segmentation Jun 2024 – Jul 2024
  – Studied **Vision-Mamba state-space models** for multi-domain medical imaging and semantic segmentation.
  – Augmented CNN-Transformer segmentation models with an adaptive learnable **fuzzy logic module** across four public datasets.
  – Raised accuracy by **2%** while reducing FLOPs, memory consumption, and parameter count by **30%**.

- **IIIT Allahabad** *Remote, Allahabad, India*
  Research Intern – Emotion Classification using EEG Mar 2024 – Jun 2024
  – Classified **24 emotions** from high-density EEG data for affective computing.
  – Optimized hyperparameters with a genetic algorithm, increasing classification accuracy and F1-score by **4%**.
  – Interpreted model outputs with **explainable AI (XAI)** to identify EEG frequency bands associated with specific emotional states.

Publications

- P. Seth*, **Anshul Kaushal***, S. Sadhu*, V. K. Sankrapu, "Drift, Then Repair: A Safety Audit of Fine-Tuned Language Models," **NeurIPS 2026 FLLMPT Workshop**. Library.
- P. Seth*, S. Sadhu*, **Anshul Kaushal***, V. K. Sankrapu, "SAFETUNE: A Unified, Faithful Library for Auditing and Repairing Safety Drift in Fine-Tuned LLMs," **EMNLP 2026 Demo Track**. Library.
- **Anshul Kaushal**, K. Jangid*, V. K. Kurmi, "CLFSeg: A Fuzzy-Logic based Solution for Boundary Clarity and Uncertainty Reduction in Medical Image Segmentation," **BMVC 2025, Core-A**. Paper.
- **Anshul Kaushal**, A. Manchanda, M. G. Dainotti*, et al., "Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves," **Journal of High Energy Astrophysics**. Paper.

- A. Manchanda, **Anshul Kaushal**, M. G. Dainotti*, et al., "Gamma-Ray Burst Light Curve Reconstruction: A Comparative Machine and Deep Learning Analysis," **The Astrophysical Journal**. Paper.
- N. Ali, M. Asif*, **Anshul Kaushal**, U. Singh, U. S. Tiwari, "Optimizing Emotion Recognition in EEG Data: A Genetic Algorithm Approach with XAI Insights," **ICCCNT 2024**. Paper.

Projects

- **Code-Grader: LLM-Assisted Grading for C Assignments**
  Program analysis, sandboxed execution, local LLMs, evaluation GitHub
  - Built a rubric-driven grader that combines **Docker-sandboxed analyzers** (compiler, cppcheck, Valgrind, ASan/UBSan, test execution) with a local **Qwen3.5-2B** model served via **llama.cpp**, producing a score out of 100 and an evidence-backed report.
  - Kept executed tests and analyzers authoritative, with the LLM rating only unmeasured metrics; multi-pass review cites evidence, then findings are deduplicated and adjudicated.
  - Made grading reproducible with temperature-0, fixed-seed decoding, bounded JSON retries, and an evidence-only fallback; generated test cases with expected outputs from reference solutions; shipped a Gradio UI and CLI.

- **COVID-19 Lung Infection Segmentation**
  Semantic segmentation, U-Net variants, medical image analysis GitHub
  - Implemented Trans, Residual, Squared, and Recurrent **U-Net** variants for COVID-19 lung infection segmentation, achieving a **Jaccard Index** $>$ **0.87**.
  - Deployed model inference with **FastAPI** and **Streamlit** for real-time predictions.

Technical Skills

- **Programming:** Python, C, C++, SQL, MATLAB, LaTeX
- **Research Areas:** LLM Safety & Alignment, Mechanistic Interpretability (circuit discovery, steering vectors), Self-Supervised & Multimodal Representation Learning, Computer Vision, Explainable AI
- **Models / Methods:** DINOv2, CLIP/MetaCLIP, Mamba SSMs, Conditional GANs, CNN-Transformers, U-Net, RAG, Genetic Algorithms, Grad-CAM
- **Frameworks / Libraries:** PyTorch, Hugging Face, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, llama.cpp, Gradio, FastAPI, Streamlit
- **Tools / Platforms:** Linux, Git, Docker, CUDA, AWS, GCP

Positions of Responsibility

- **Core Team,** Training and Placement Cell, Panjab University *Oct. 2023 – 2026*
  - Collaborated in a **15+ member team** to organize **10+ workshops and events** on machine learning and AI, impacting **200+ students**.

Achievements

- MEXT 2027 Scholarship – top 186 candidates from 7,300 applicants (top 2.5%)
- ACM IKDD Uplink Research Internship 2025 – 2% acceptance rate
- Oxford Machine Learning Summer School (OxML) 2025 – shortlisted
- IELTS Academic – 8.0 Overall Band

Certifications

- MLx Fundamentals (OxML): Machine Learning, Deep Learning, Representation Learning, Optimization, Generative AI
- DeepLearning.AI (Coursera): Unsupervised Learning, Recommender Systems, Reinforcement Learning
- Imperial College London (Coursera): Mathematics for Machine Learning – Linear Algebra
- Amazon Web Services: AWS Elements of Data Science
- Duke University (Coursera): Data Science Math Skills